HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and refined **ML models** for classification and ranking tasks, focusing on improving model accuracy, precision, and latency for real-time user interactions.
- Built and maintained **ML training, inference, and deployment pipelines** on AWS, ensuring reproducibility and maintainability across different model versions.
- Engineered **Python data pipelines** to process and transform large volumes of user interaction data, preparing it for model training and evaluation.
- Implemented **retrieval and personalization models** that learn from user behavior, incorporating techniques for hybrid search and reranking to enhance search relevance.
- Designed and deployed **production ML systems**, debugging issues in live environments and monitoring model performance to drive continuous improvements based on real-world feedback.
- Developed robust **evaluation pipelines** and curated labeled datasets with precise metrics to measure model quality and identify regressions before deployment.
- Containerized **ML inference services** using Docker and deployed them to a Kubernetes cluster, managing configurations and resource allocation for predictable scaling.

SKILLS

ML Core – Classification, Retrieval, Ranking, Personalization, Model Evaluation, Neural Architectures, Fine-tuning, Embeddings, Vector Search

MLOps & Deployment – Production ML Systems, Inference Pipelines, Training Pipelines, Deployment Pipelines, Docker, Kubernetes, AWS

Programming & Data – Python, PyTorch, JAX, SQL, pandas, NumPy

Model Development – Classifier Optimization, Accuracy, Precision, Latency, Cost, Reliability, Experimentation, Labeled Datasets, Metrics

Cloud & Infrastructure – AWS, EC2, S3, Lambda, CloudWatch, Linux, Git, Jenkins

PROJECTS

Personalized Content Recommendation Engine

- Developed a **Python-based recommendation engine** that analyzes user interaction data to provide personalized content suggestions, improving user engagement by approximately 15%.
- Trained and fine-tuned a **content-based filtering model** using scikit-learn on a dataset of 1.5 million user-item interactions, achieving a Mean Reciprocal Rank (MRR) of 0.72.
- Built RESTful APIs using **FastAPI** to serve real-time recommendations, integrating with an existing web application and ensuring p95 latency below 200ms.

ML System for User Intent Classification

- Engineered a **classification pipeline** in Python to identify user intent from textual inputs, processing an average of 5,000 requests per day.
- Implemented a **PyTorch-based neural network** for text classification, achieving 91% accuracy on a held-out test set of user queries.
- Developed an **evaluation framework** to track classification performance, including precision, recall, and F1-score, deployed on AWS Lambda for automated testing.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)